

# SWINBURNE UNIVERSITY OF TECHNOLOGY

## Professional Placement (3-months)

### Assignment 2 – Intention for Learning on Placement

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For this assignment, please complete the requirement checklist below as per the instructions on how to prepare and complete Assignment 2 on Canvas.

| Requirement Checklist                                                                                                     | Date of the meeting | Tick to confirm                     |
|---------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------|
| I have met and discussed the assignment draft with my Academic Supervisor.                                                | 27/05/2026          | <input checked="" type="checkbox"/> |
| I have discussed my intentions for learning in the Placement meeting with my Industry Supervisor and Academic Supervisor. | 03/06/2026          | <input checked="" type="checkbox"/> |
| I have had or are planning my second meeting with my peer-to-peer (P2P) debrief group.                                    | 18/05/2026          | <input checked="" type="checkbox"/> |

#### SECTION 1: Intention for Learning on Placement (approximately 600 words in total)

- In this section, you are required to reflect on three out of four domains of practice below. The **Knowledge in Practice** domain is mandatory. Please tick and **choose 2 of the remaining domains of practice** to identify your specific learning intentions whilst on placement.

☒ **Knowledge in Practice (required)**

☒ Practice of Work

☒ Ethical Work

☒ Career Development Practice

- Identify the specific learning focus for **Domain 1 – Knowledge in Practice** and 2 other of your chosen domains below. For each domain, provide a brief description of 200 words referring to the reflective prompts:
  - What?** - Name and briefly describe the disciplinary knowledge you intend to focus your learning on for this domain (e.g. areas, frameworks, concepts, theories, methodologies,...).
  - So what?** - Explain why this focus is relevant or important to you.
  - Now what?** - Identify three actions you will take to develop your capabilities in this domain.

Use your answers in the WIL Capabilities Self-assessment (Assignment 1) to guide you.

| Knowledge in Practice (required)                                                                                                                                                                                                                                                                                                                                                                                                     |
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| What?                                                                                                                                                                                                                                                                                                                                                                                                                                |
| My learning focus is on Cloud-Native Architecture and Serverless Microservices. Specifically, this involves mastering the integration of a managed Backend-as-a-Service (Supabase) with an event-driven AWS serverless ecosystem (Lambda, EventBridge, S3, API Gateway). It also encompasses the implementation of an offline-first client architecture using Flutter and Drift (SQLite) to ensure zero-disruption user experiences. |

| So what?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Mastering this hybrid, distributed infrastructure is vital for developing scalable, highly available applications without relying on traditional, cost-heavy monolithic architectures. It bridges the gap between theoretical systems engineering and practical, zero-budget cloud deployment. Understanding how to orchestrate local data queues with remote cloud services ensures robust data synchronization and fault tolerance, which are critical competencies for a modern data and cloud engineer. |
| Now what?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Design and implement a local sync queue manager in Flutter to handle seamless offline-to-online data transitions and resolve synchronization conflicts.<br><br>Deploy a Python-based Natural Language Processing (NLP) microservice on AWS Lambda to process text-based emotion detection natively in the cloud.<br><br>Configure automated serverless cron jobs via Amazon EventBridge to handle background analytics and generate markdown reports.                                                       |

| 2 <sup>nd</sup> Domain: Innovation and Problem Solving                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| What?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| The focus here is on Applied Artificial Intelligence and Retrieval-Augmented Generation (RAG) Systems. This involves leveraging vector databases (pgvector within PostgreSQL) and Generative AI (such as Amazon Bedrock) to analyze user emotions and behavioral patterns, transforming static log data into personalized, proactive digital interventions.                                                                                                                                  |
| So what?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Modern software must move beyond passive data collection. Traditional productivity tools track time but ignore the user's psychological state. By innovating an intelligent system that infers emotional exhaustion from text and dynamically curates contextual self-help content (e.g., therapeutic quotes or schedule adjustments), the application actively solves complex user challenges. This elevates the project from a standard utility to an empathetic, proactive digital coach. |
| Now what?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Configure and optimize pgvector to store and execute similarity searches on text embeddings (e.g., transcripts, quotes, and media links).<br>Build a secure API communication layer allowing the AWS Lambda recommender function to query the self-hosted PostgreSQL database.<br>Develop the algorithmic logic for the AI Agent to analyze historical focus sessions and trigger real-time health break notifications via Amazon SNS.                                                       |

| 3 <sup>rd</sup> Domain: Self-Management and Professional Practice |
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| <b>What?</b>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| My focus is on Agile Project Management and Continuous Integration/Continuous Deployment (CI/CD) Operations. This entails executing a highly structured, iterative development plan—moving away from rigid waterfall documentation toward Minimum Viable Product (MVP) sprints, automated testing, and streamlined release pipelines using Github.                                                                              |
| <b>So what?</b>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Successfully delivering a complex, multi-service application requires strict discipline and operational efficiency. Prioritizing an MVP ensures development momentum is maintained and core features are validated early. Furthermore, implementing automated CI/CD pipelines ensures reliable code quality and prepares me for the rigorous operational and DevOps standards expected in professional enterprise environments. |
| <b>Now what?</b>                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>Complete the foundational phase by finalizing UI/UX wireframes and the Entity-Relationship Diagram (ERD) before initiating the first coding sprint.</p> <p>Maintain a detailed Weekly Project Log to systematically document technical roadblocks and problem-solving strategies.</p> <p>Write targeted unit tests and establish a GitHub CI/CD pipeline to automate code validation.</p>                                    |

## SECTION 2: Point of Professional Inquiry (approximately 500 words)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Briefly describe your current professional interests and / or your sense of professional purpose in relation to ONE of the UN Sustainability Goals that connects with your interests and explain how they connect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| My professional purpose lies at the intersection of cloud engineering, data analytics, and user-centric software development. I am deeply interested in leveraging distributed cloud architectures and artificial intelligence to build systems that are not only highly performant but also actively improve the daily lives of their users. This aligns directly with <b>UN Sustainability Goal 3: Good Health and Well-being</b> . While technology has rapidly accelerated our ability to work and communicate, it has also introduced significant cognitive burdens. My interest is in utilizing technology to reverse this trend—building intelligent applications that promote mental well-being, manage cognitive load, and foster sustainable work habits.                             |
| Identify and briefly describe a challenge facing your industry/professional field that relates to both your professional interests and the identified UN Sustainability Goal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| A critical challenge in the modern digital workplace—and the software industry itself—is the epidemic of burnout and digital fatigue. The expectation of constant connectivity and context-switching fractures focus and degrades mental health. Currently, most productivity software (such as standard task managers or Pomodoro timers) operates mechanically; they measure efficiency through time elapsed or tasks checked off, entirely ignoring the emotional and physical state of the human using them. The industry lacks proactive, empathetic tools that can identify when a user is experiencing mental exhaustion and intervene before burnout occurs.                                                                                                                            |
| Describe how this might connect with your placement – e.g. work, organisation, industry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| My current project, the development of an Event-Driven Full-Cloud Focus Mode App, is a direct response to this challenge. By moving away from a standard timer to an "immersive" offline-first environment, the application structurally minimizes digital distractions. More importantly, it actively connects to mental well-being through its AI-driven features. By utilizing a Python NLP microservice to detect emotional states from post-session journal entries, and a RAG system to deliver personalized, grounding content (such as balanced scheduling advice or restorative quotes), the application acts as a digital productivity coach. It tracks the intersection of efficiency and mental health, stepping in with real-time health breaks when prolonged strain is detected. |
| Describe some ideas or a question you might now like to explore during your placement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| During the development and testing of this application, I want to explore the following question: <i>How can we effectively balance edge computing (local offline processing) with cloud-native AI services to deliver</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

*real-time, privacy-preserving mental health interventions without compromising system performance?*  
Additionally, I want to investigate how accurately a lightweight NLP model can track subtle shifts in a user's productivity patterns over time, and whether an AI Agent can reliably suggest automated schedule adjustments that users actually find beneficial for their mental well-being.

### **SECTION 3: References**

#### **Resubmission acknowledgement**

This section is not required if this is your first submission.

If your Academic Supervisor requires you to resubmit for a reassessment, please tick the box below to acknowledge that you have understood the requirements.

☒ I have addressed and incorporated the feedback received. I am aware that failing to do this could result in a fail for this assignment and would have, this has implications to passing the unit (refer to unit outline).